Ice Stupa Fellowship (FAQs)

About the Fellowship

1. What is the Ice Stupa Fellowship?

The Ice Stupa Fellowship is an interdisciplinary research and innovation programme organised by the Himalayan Institute of Alternatives, Ladakh (HIAL). It brings together students from engineering, physical sciences, environmental sciences, architecture, and related fields to explore technological solutions for water-related challenges in high-altitude Himalayan environments. It moves beyond classroom to experiential teaching & learning models with real life problems.

2. What is the main focus of the fellowship?

The fellowship focuses on Ice Stupa technology, artificial ice systems, mountain hydrology, and associated engineering and scientific challenges. It also explores how technologies such as sensors, automation, computational modelling, and new materials can contribute to the design, monitoring, and improvement of water systems in mountain environments.

3. What is an Ice Stupa?

An Ice Stupa is an artificial ice structure that stores water in frozen form during winter and gradually releases it as the ice melts in spring. In suitable locations, it can help make water available during periods of seasonal water scarcity.

4. Why is this fellowship being conducted in Ladakh?

Ladakh's high-altitude cold-desert environment presents distinctive challenges related to water availability, freezing temperatures, seasonal melting, and difficult terrain. It provides a valuable setting for investigating how engineering and scientific solutions can be designed and tested under challenging environmental conditions. Once the fellow is equipped with all the technological and field expertise, one can become partners with HIAL in their own region/location/countries.

Eligibility and Applications

5. Who can apply for the fellowship?

People only from **Himalayan region in India** and **Nepal & Bhutan** above the age of 23 to 30 with relevant academic backgrounds are eligible to apply, including those studying:

- Civil Engineering and Architecture
- Mechanical Engineering and Materials Science
- Electronics and Electrical Engineering
- Computer Science, IoT, and Automation
- Physics
- Glaciology and Hydrology
- Environmental Science and Climate Science

We are also open to those people beyond the above background to apply but having deep interest and keen in working with Artificial Glaciers systems in their own regions to help community facing water scarcity.

6. Can students from different disciplines work together?

Yes. Interdisciplinary collaboration is central to the fellowship. Many challenges associated with Ice Stupa technology require knowledge from multiple fields. For example, a project involving automated monitoring may combine mechanical engineering, electronics, computer science, and hydrology.

7. Do I need prior experience with Ice Stupa technology?

Prior experience with Ice Stupa technology is not necessarily required. An interest in the fellowship's themes, a willingness to learn, and the ability to engage with practical research problems, adaptive to work in cold weather are valuable.

Research and Learning Experience

8. What will I do during the fellowship?

Fellows will work on field-oriented research and engineering problems, develop and test technological solutions, collaborate with scholars and researchers, and participate in prototyping, experimentation, modelling, and field studies.

9. What research themes can I explore?

The fellowship covers eight broad themes:

1. Ice Stupa and Artificial Ice Systems
2. Mountain Hydrology
3. Automation and Autonomous Technologies
4. IoT, Sensing and Wireless Systems
5. Mechanical and Materials Engineering
6. Structural Design of Artificial Glaciers
7. Modelling, Simulation and Algorithms
8. Glaciology and Cryosphere Research

These themes offer different entry points into the wider challenges of artificial ice systems and water management in the Himalayas.

10. Will the fellowship involve practical, hands-on work?

Yes. Practical engagement is an important and central part of the HIAL as well as the fellowship's approach. Depending on the project, this may involve field work, experimental work, basing in villages, prototype development, testing, simulation, data collection, and analysis of engineering problems.

11. Will I have opportunities to work on my own research ideas?

The fellowship encourages curiosity, problem-solving, and innovative thinking. Opportunities to develop individual ideas or contribute to collaborative projects will depend on the programme structure, available resources, and alignment with the fellowship's research priorities.

12. Will I work with researchers from other disciplines?

Yes. Fellows are expected to engage in interdisciplinary collaboration with researchers, scholars, and participants from different technical backgrounds. This approach helps bring different perspectives and expertise to complex research problems.

13. Will I learn about technologies beyond Ice Stupas?

Yes. Although Ice Stupa technology provides the central context, the fellowship also explores related areas such as hydrology, sensing systems, automation, materials engineering, structural design, computational modelling, and cryosphere research.

Practical Information

14. Where will the fellowship take place?

The fellowship will take place in HIAL campus Phyang Leh. During the course of fellowships you will be stationed to various villages in Ladakh to conduct field research and experimentation.

15. What is the duration of the fellowship?

The duration of the fellowship is Six Months, especially in winter months from Nov to April.

16. Is the fellowship residential?

Yes, the fellowship is fully residential. While in campus you will be provided passive solar rooms and in the field, homestays in the villages will be your home away from home.

17. Is there any Fellowship fee or course fee?

The fellowship is free of cost, there is no course fee as such. However, you have to cover your food and accommodation with 10,000 per month.

18. What is the application process?

Applicants should complete the official application form and provide the information and documents requested in the google form.

19. How will applicants be selected?

The selection criteria will be based on their academic background, interest in working with artificial Ice systems.

20. How cold it gets in winter in Ladakh?

Winter in Ladakh remains very cold temperature ranging from -20 to 10 degree Celsius.

21. What special items I should carry to Ladakh.

Well, the more you prepare the well you survive. Since you will have field visits and sometimes stays in villages. You need to carry sleeping bags good enough for minus temperature. A pair of hiking or trekking shoe durable enough for rugged terrain. At least two pairs of thermals/warmer, trousers and a hiking backpack. You will also need warm caps, socks and thermos/flask to keep your warm water. These items may cover a lot of baggage quota during your flights, you can also buy all these items in the local markets.